Vishesh R S
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Executive Summary

Motivated Mechanical Engineering student interested in 3D CAD modelling, product design, and emerging technologies. Proficient in SOLIDWORKS, with hands-on learning in sketching, part modelling, assemblies, and advanced design features. Interested in integrating mechanical engineering with AI, robotics, and sustainable technology to develop practical solutions.

Technical Skills

CAD & 3D Modeling: SOLIDWORKS (Parametric Modeling, Tolerancing), Solid Edge, Fusion 360, Web-Based 3D Visualization (React, Three.js)

Simulation & Analysis: ANSYS Fluent (CFD), ANSYS Static Structural (FEA), Kinematic and Dynamic Motion Analysis

Software & Programming: Python, HTML, CSS, Machine Vision (YOLOv8)

Multidisciplinary Technology: IoT Telemetry, Digital Twin Modeling, Embedded Systems, Relay-Based Automation

Soft Skills

- Problem Solving & Critical Thinking
- Cross-Functional Teamwork
- Adaptability in Multidisciplinary Projects
- Technical Communication
- Time Management

Professional Experience

1) Smart India Hackathon 2026 (JALPRAGYA) 2026

August 2026 – Present

Role: Multidisciplinary Project Lead & CAD Designer

Responsibilities:

- Formed and lead a multidisciplinary team developing a cost-effective water purification and quality-monitoring solution for rural and mining-affected regions of Jharkhand.
- Contribute to the proposed system architecture, combining multi-stage filtration, reverse osmosis (RO), and ultraviolet (UV) disinfection.
- Work with a monitoring concept incorporating pH, total dissolved solids (TDS), turbidity, and temperature sensors with an ESP32-based system.
- Focus on integrating purification stages and real-time water-quality information into a practical prototype.

Impact: The project aims to support safer drinking-water decisions by combining purification with water-quality monitoring; prototype validation and field performance are not claimed.

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2) MSME Idea Hackathon 6.0 (Omni-Node) 2026

Role: Hardware Architecture Contributor

Responsibilities:

- Contributed to an AI-driven retrofit ecosystem concept intended to improve flexibility in High-Mix Low-Volume (HMLV) manufacturing.
- Developed hardware component architecture and detailed system block diagrams.
- Prepared visual presentation frameworks to communicate the proposed hardware–software integration concept.

Impact: Formulated a proposed retrofit framework connecting modern AI software concepts with legacy manufacturing hardware.

3) 4-Cylinder Engine Assembly **2026**

Role: Mechanical Design Contributor

Responsibilities:

- Designed a parametric 3D assembly of a four-cylinder internal combustion engine using SOLIDWORKS.
- Modelled major components, including pistons, crankshaft, connecting rods, cylinder block, and valvetrain.
- Applied dimensional and tolerance considerations to mating components and performed assembly interference checks.
- Simulated the dynamic motion of the piston–crankshaft mechanism under simulated operating loads.

Impact: Evaluated kinematic behaviour and component fit through CAD assembly and motion analysis before physical prototyping.

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4) Offline Voice-Controlled Switch **2025 - 2026**

Role: Idea Pitching and Hardware Lead

Responsibilities:

- Designed a localized voice-controlled switching system using the AI-Thinker VC-02 offline speech-recognition module.
- Integrated a four-channel relay module to control multiple electrical components independently through voice commands.
- Developed an automation approach that does not rely on cloud-based speech processing or an internet connection.

Impact: Developed an offline voice-command switching prototype without dependence on cloud processing.

Education

Bachelor of Engineering (B.E.), Mechanical Engineering

August 2025 – Present
Sri Krishna College of Engineering and Technology (SKCET), Coimbatore

Awards & Accomplishments

- Competitor & Team Lead, Smart India Hackathon 2026 (JALPRAGYA & ANVESHA).
- Competitor & Team Lead, Seva Innovation Challenge 2026 (JALPRAGYA).
- Competitor, MSME Idea Hackathon 6.0 (Omni-Node).
- Preparing for the Certified SOLIDWORKS Professional (CSWP) examination.